

Go to tutorial!!! HW1 is due sunday at 8pm. Mathematics SUpport Initiative, see Quercus, math help club.

Recap. Recall that $\mathbb{Z}/m\mathbb{Z} = \{\bar{0}, \dots, \overline{m-1}\}$ is the set of equivalence classes mod m , equipped with the operations $+$ and $\cdot$, given by:

1. Do the operation as usual
2. Then, take the remainder

Remark. What is a prime number?

Definition 1. Let $p \in \mathbb{N}$, $p > 1$, then p is prime if $\forall x, y \in \mathbb{Z}$,

$$p \nmid x \text{ and } p \nmid y \implies p \nmid xy$$

asfasf

Remark. What is a prime number?

Check this as an exercise! This is also called Euclid's lemma.

Theorem 2. Let $m \in \mathbb{N}$. If m is prime, then $\mathbb{Z}/m\mathbb{Z}$ is a field. blahblahblahblahblahblah
blah blah blah blah

Remark. What is a prime number?

Proof. First, check the field axioms, with (F4) holding for addition. ■